

# Adams fixed points on a cyclic group

## Disproof of Conjecture 00000008230

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**Result.** For the cyclic group  $C_3$  and the second Adams operation, the fixed subspace of complex class functions has dimension 2, whereas the number of 2-regular conjugacy classes is 3. Thus the dimension equality asserted in the conjecture is false.

### 1. Definitions and the disputed assertion

For a finite group  $G$ , write  $\text{CF}(G; \mathbb{C})$  for its vector space of complex class functions. These are functions  $f : G \rightarrow \mathbb{C}$  such that  $f(xgx^{-1}) = f(g)$  for every  $x, g \in G$ . The Adams operation on class functions is

$$(\psi^k f)(g) = f(g^k).$$

Its fixed subspace consists of the class functions satisfying  $f(g^k) = f(g)$  for all  $g$ . An element is  $k$ -regular when its order is coprime to  $k$ ; a conjugacy class is  $k$ -regular when its elements are. For  $k = 2$ , this is simply the familiar odd-order condition.

The English statement explicitly asserts that the dimension of this fixed subspace equals the number of  $k$ -regular classes. The Chinese statement asserts the same dimension equality. We disprove this conjunct. The conjecture's other assertions about spectral support and composition corrections are not needed for the contradiction.

### 2. The counterexample

Let

$$G = C_3 = \langle a \mid a^3 = 1 \rangle = \{1, a, a^2\}, \quad k = 2.$$

The group is abelian, so its three conjugacy classes are the singletons  $\{1\}$ ,  $\{a\}$ , and  $\{a^2\}$ . Their element orders are 1, 3, and 3, respectively. All three are coprime to 2. Consequently,

$$\#\{2\text{-regular conjugacy classes of } C_3\} = 3.$$

Every function on  $C_3$  is a class function. Writing its values in the order  $(1, a, a^2)$ , the second Adams operation is

$$\psi^2(u, v, w) = (u, w, v),$$

since  $1^2 = 1$ ,  $a^2 = a^2$ , and  $(a^2)^2 = a^4 = a$ .

Thus  $\psi^2 f = f$  holds exactly when  $f(a) = f(a^2)$ . Its fixed subspace is

$$F = \{(u, v, v) : u, v \in \mathbb{C}\}.$$

The maps

$$F \longrightarrow \mathbb{C}^2, \quad f \longmapsto (f(1), f(a)), \quad (u, v) \longmapsto [g \mapsto \begin{cases} u, & g = 1, \\ v, & g \neq 1 \end{cases}]$$

are inverse complex linear maps. Therefore  $\dim_{\mathbb{C}} F = 2$ . Equivalently, the functions with values  $(1, 0, 0)$  and  $(0, 1, 1)$  form a basis. Combining the two exact computations gives

$$\dim_{\mathbb{C}} \ker(\psi^2 - I) = 2 \neq 3 = \#\{2\text{-regular conjugacy classes}\}.$$

This disproves the stated equality and hence the conjunctive conjecture.  $\square$

The reason the proposed count fails is already visible here: squaring permutes the two non-identity conjugacy classes, so a fixed function must take equal values on them. Counting the classes individually does not count the independent values of a fixed function.

### 3. Lean correspondence and verification

The Lean project uses Lean 4.19.0 and pins Mathlib to commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The finite group is `Multiplicative (ZMod 3)`, with its actual group operations. Conjugacy classes are Mathlib’s quotient `ConjClasses G`; regularity uses `orderOf` and `Nat.Coprime`. In particular, neither the number of classes nor the element orders are supplied as unproved data.

`cube_one` checks the group identity  $g^3 = 1$  for every element. It follows that each order divides 3, proving regularity. `regular_class_count` computes the cardinality of the actual subtype of regular conjugacy classes, using the conjugacy-class bijection for a commutative group.

`fixedClassFunctions` is a complex submodule whose defining conditions are conjugation invariance and the equation  $\psi^2 f = f$ . The linear equivalence `coordinates` implements the two inverse maps above and proves both inverse identities and linearity. `fixed_dimension` then obtains dimension 2 through this equivalence, rather than assuming a basis or a dimension table.

The final theorem `Adams8230.counterexample` states that the two actual quantities are unequal. Reproduction commands and the axiom audit are in `VERIFICATION.md`. There are no additional axioms, incomplete proofs, or `native_decide`; no auxiliary computational program is required.